

CampusEats — Assignment 2

Services, Contracts & Database Design

Team Members

Anurag Sharma — Leader

Mohit Tailor

Harshita Rathore

Harendra Godara

1. Capabilities

The capabilities are derived from the CampusEats Assignment 1 brief: students discover food, place and track orders; food providers manage menus and incoming orders; administrators manage the overall system.

1	Register and authenticate users
2	Manage users, food providers and administrator access
3	Browse and search available food
4	Manage menus and food item availability
5	Create and place food orders
6	Process and authorize payments
7	Accept or reject incoming orders
8	Prepare orders and update order status
9	Track order status
10	View previous orders
11	Select pickup location
12	Monitor and maintain system information

2. Services, Ownership & Team Members

Service	Responsible Team Member	Data Owned
Identity & Administration Service	Anurag Sharma (Leader)	• Users • Roles • Food Provider profiles • Administrator access
Catalogue Service	Mohit Tailor	• Restaurants • Menus • Food Items • Categories • Availability • Pickup Locations
Order Service	Harshita Rathore	• Orders • Order Items • Order Status • Order Pickup selections
Payment Service	Harendra Godara	• Payments • Payment Status • Payment Transactions

Cross-service calls: Order → Catalogue: checkItems; Order → Payment: authorizePayment; Identity → Order: verifyAccess.

3. Service Contracts

Each service exposes operations through a clear contract. Internal database implementation details are hidden from callers.

Identity & Administration Service — Anurag Sharma (Leader)

Operation	Input	Output	Errors
getUser	userId	User profile + role	User not found
registerUser	name, email, role	userId, account status	Invalid data, duplicate account
verifyAccess	userId, requiredRole	allowed	User not found, access denied

Catalogue Service — Mohit Tailor

Operation	Input	Output	Errors
searchItems	query, category, campus	matching food items	Invalid search criteria
getItem	itemId	item details + availability	Item not found
checkItems	items[{itemId, quantity}]	validated item list + current prices	Item unavailable, insufficient quantity
updateAvailability	itemId, available	updated availability	Item not found, not authorized

Order Service — Harshita Rathore

Operation	Input	Output	Errors
createOrder	studentId, items, pickupLocation	orderId, status, total	Invalid items, unavailable item
getOrder	orderId, requesterId	order details + status	Order not found, access denied
updateStatus	orderId, providerId, status	updated order status	Invalid transition, access denied
listStudentOrders	studentId	order history	Student not found

Payment Service — Harendra Godara

Operation	Input	Output	Errors
authorizePayment	orderId, amount, paymentMethod	paymentId, status	Payment declined, invalid amount
getPayment	paymentId	payment status + transaction details	Payment not found
refundPayment	paymentId, amount	refund status	Payment not refundable

4. Central Operation — placeOrder

Aspect	Specification
Purpose	Create a student food order after validating requested items and pickup information.
Inputs	studentId; items[{itemId, quantity}]; pickupLocationId; paymentMethod when payment is required.
Outputs	orderId; order status; confirmed order items; total amount; payment status when applicable.
Success	Order Service creates the order and returns the placed order reference and total.
Error: unavailable item	If Catalogue reports an item unavailable or quantity cannot be fulfilled, no order is placed.
Error: invalid item	If an item reference is invalid, the operation fails with a validation error.
Error: access denied	If the caller is not allowed to place an order for the student, the operation fails.
Error: payment declined	If payment is required and Payment Service declines it, the order is not confirmed as paid.
Error: invalid pickup	If the pickup location is invalid or unavailable, the operation fails.
Service calls	Order → Catalogue: checkItems. Order → Payment: authorizePayment. Order → Identity: verifyAccess when authorization is required.
Hidden internals	Callers do not see order tables, SQL, transaction handling, price calculation implementation, payment-provider details, or other implementation details.

5. Service Validation

Validation properties: reachable, self-contained, has a contract, independent, and loosely coupled.

Service	Reachable	Self-contained	Contract	Independent	Loosely coupled	Result
Identity & Administration Service	Yes	Yes	Yes	Yes	Yes	Pass
Catalogue Service	Yes	Yes	Yes	Yes	Yes	Pass
Order Service	Yes	Yes	Yes	Yes	Yes	Pass
Payment Service	Yes	Yes	Yes	Yes	Yes	Pass

6. Database Design Summary

Identity owns users and provider profiles. Catalogue owns restaurants, menus, categories, food items and pickup locations. Order owns orders and order items. Payment owns payments. The editable ER diagram is supplied as schema.drawio and the SQL CREATE TABLE sketch as schema.sql.